

## Review

# Large language models to process, analyze, and synthesize biomedical texts: a scoping review

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Received: 7 August 2024 / Accepted: 7 November 2024

Published online: 19 December 2024

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## Abstract

The advent of large language models (LLMs) such as BERT and, more recently, GPT, is transforming our approach of analyzing and understanding biomedical texts. To stay informed about the latest advancements in this area, there is a need for up-to-date summaries on the role of LLM in Natural Language Processing (NLP) of biomedical texts. Thus, this scoping review aims to provide a detailed overview of the current state of biomedical NLP research and its applications, with a special focus on the evolving role of LLMs. We conducted a systematic search of PubMed, EMBASE, and Google Scholar for studies and conference proceedings published from 2017 to December 19, 2023, that develop or utilize LLMs for NLP tasks in biomedicine. We evaluated the risk of bias in these studies using a 3-item checklist. From 13,823 references, we selected 199 publications and conference proceedings for our review. LLMs are being applied to a wide array of tasks in the biomedical field, including knowledge management, text mining, drug discovery, and evidence synthesis. Prominent among these tasks are text classification, relation extraction, and named entity recognition. Although BERT-based models remain prevalent, the use of GPT-based models has substantially increased since 2023. We conclude that, despite offering opportunities to manage the growing volume of biomedical data, LLMs also present challenges, particularly in clinical medicine and evidence synthesis, such as issues with transparency and privacy concerns.

**Keywords** Natural language processing · Bioinformatics · Biomedicine · Large language models · BERT · Evidence synthesis

## 1 Introduction

Natural Language Processing (NLP), a field at the intersection of computer science, artificial intelligence and linguistics, aims at enabling computers with the ability to comprehend language similarly to how humans do. It has become an indispensable tool for processing and analyzing large volumes of text data, widely adopted in various fields including biomedical research and clinical applications [1]. A pivotal methodological advancement in NLP has been the transformer architecture, which underpins most recent Large Language Models (LLMs), including both BERT, Bidirectional Encoder

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**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s44163-024-00197-2>.

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Representations from Transformers, and GPT, Generative Pre-trained Transformer. The transformer model is appreciated for its efficiency and ability to process data in parallel. This capability, combined with a deep learning architecture that leverages attention mechanisms, allows for handling long dependencies and yields a better contextual understanding [2]. Such advancements have fundamentally altered the landscape of NLP model development. The focus has shifted towards refining and customizing pre-trained language models, moving away from the traditional approach of building models from the ground up. This approach is especially advantageous in specialized fields like biomedicine, where annotated datasets are often limited or difficult to compile. By leveraging the fine-tuning capabilities of LLMs for specific tasks, researchers in the biomedical domain can attain high-performance levels, even when working with relatively small datasets [3].

Amidst these advancements, the emergence of generative models marks an evolution towards creative and generative applications, setting a contrast to the discriminative nature of earlier models [4, 5]. While BERT and similar systems excel in task-oriented applications such as sentence classification or entity recognition, generative models are capable of producing coherent and contextually relevant text based on the input they receive. This capability facilitates novel applications such as automated content creation, summarization, and question-answering systems in the biomedical domain. In addition, these models show promising capabilities in exploring instruction tuning and in-context learning. This enables them to comprehend and execute a wide array of tasks using natural language instructions or examples, thus further reducing the reliance on task-specific datasets or extensive fine-tuning [6]. However, generative models also introduce new risks and challenges, such as the generation of plausible but incorrect or misleading information, underscoring the need for careful evaluation and application of these technologies [7, 8].

Given the fast-paced nature of NLP development, there is a need for up-to-date literature overviews. This is particularly true in fields like biomedical research, where the ability to swiftly and accurately process vast amounts of text data can substantially impact patient care and scientific discovery [9]. Thus, this scoping review has three goals: Firstly, we aim to identify and categorize the specific tasks within BioNLP that are currently being addressed using LLMs, e.g., in evidence synthesis, pharmacology, or clinical use cases. Secondly, we aim to map the various LLM architectures employed in these BioNLP tasks. Finally, this study sets out to assess the transparency of methods used in the development of these LLMs, including the availability of source code and data as well as used hardware and software.

## 1.1 Related work

The work presented in [10] presented the first comprehensive review of the development and landscape of transformer-based biomedical language models. It examined the core implementation principles behind these models and their practical applications for BioNLP tasks with a focus on encoder-based language models. Subsequent work expanded the discussion to include RNN-based models and those that leverage decoders for generative pre-training [3]. Furthermore, the authors extended their exploration of biomedical LLMs to encompass applications beyond textual data, including analyses of proteins, DNA, and text-image pairs. Aligning with our goals, they also provided an in-depth exploration of language model applications within the biomedical domain, offering a thorough classification of techniques, datasets, and benchmarking competitions [3].

Other reviews have concentrated on the use of LLMs within specific niches of biomedical text processing. For instance, how LLMs can enhance downstream tasks in literature-based discovery [11]. Another survey shed light on biomedical and clinical NLP research conducted in languages other than English, focusing on data resources, language models, and common NLP tasks in these languages. This survey also highlighted the growing trend of employing transformer-based language models for various NLP tasks in medical fields [12].

Recent surveys have delved into the unique challenges posed by deploying LLMs in healthcare, especially concerning fairness and accountability. For example, an overview of the technical aspects of development and usage of LLM methods in healthcare, including the specific challenges such as ethical considerations encountered when deploying this technology in the field, has been provided [13]. Similarly, the potential of models such as ChatGPT to enhance medical practices, while addressing concerns about safety, ethics, and maintaining the human element in patient care, has been reviewed in [7]. A related systematic review evaluated the benefits and limitations of ChatGPT in healthcare education, research, and practice, based on an analysis of 60 publications from PubMed/MEDLINE and Google Scholar [14]. Insights into the opportunities and challenges of LLMs for biomedical or clinical usage, including their roles in pre-consultation, diagnosis, data management, and their support for medical education and writing, have also been offered in [15].

In this scoping review, we systematically examine the usage of both discriminative and generative LLM architectures in biomedical text-processing applications. Unlike existing reviews that often focus on specific applications or model

architectures using narrative approaches, our study adopts a rigorous systematic review methodology to identify and evaluate the current state of research to mitigate biases. Through in-depth quantitative analyses, we uncover temporal trends and patterns that highlight the evolving landscape of biomedical NLP. Notably, to the best of our knowledge, we are the first to quantify and critically assess transparency and Open Science practices within the BioNLP community by evaluating aspects such as data availability, code accessibility, and methodological transparency.

## 2 Methods

### 2.1 Study registration

We registered the study protocol on the Open Science Framework (OSF) platform (<https://osf.io/bjv24/>).

### 2.2 Literature search

We conducted a search for studies in PubMed and EMBASE, spanning from 2017 (i.e., the inception of transformer-based LLMs [2]) to December 19, 2023. Additionally, we searched Google Scholar for EMNLP and ACL machine learning conference proceedings through the “Publish or Perish” software, a tool designed for retrieving and analyzing academic citations from multiple databases<sup>1</sup>. This software was necessary because Google Scholar does not enable downloading of larger chunks of references. It also provides an easy method for downloading metadata and paper information of search results. We employed a search string translated from our PubMed query to ensure consistency across data sources, which was created by an information specialist at the University Library Zurich. The complete search string can be found in the Appendix.

### 2.3 Inclusion and exclusion criteria

We included original research articles that employed LLMs to analyze extensive collections of biomedical texts, e.g., scientific publications, (pre)clinical trial registries, patents, and grey literature. Additionally, systematic reviews and meta-analyses that leveraged LLMs for automating certain tasks like abstract screening were also considered.

We excluded non-English publications as well as text data derived from clinical questionnaires or surveys. Non-systematic reviews were also excluded but were retained as a supplementary source for additional references.

### 2.4 Study selection

All retrieved publications were screened using ASReview, which incorporates machine learning algorithms to prioritize relevant studies [16]. We established a rule in advance to cease the screening of abstracts once we encountered thirty consecutive irrelevant abstracts. Following this, any remaining abstracts were excluded from further analysis.

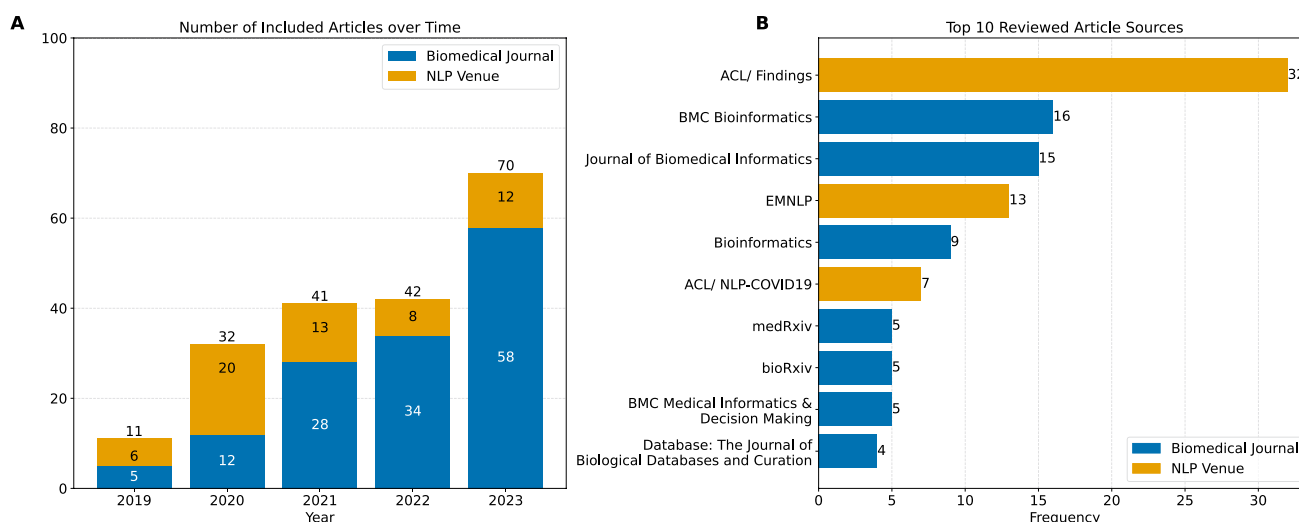
### 2.5 Data extraction and synthesis

The following data were extracted from available full texts of eligible articles: target NLP application (grouped into pre-specified classes), biomedical domain of the application, target database for the application of the developed methods (e.g., PubMed), data type (e.g., abstracts), data availability, hosted application for end-users, LLM (e.g., BioBERT), programming language (e.g., Python), library/framework (e.g., HuggingFace), reported performance metrics (e.g., F1-score), source code availability, pretraining corpus origin (if pre-training was performed, which data was used, e.g., COVID-19 Open Research Dataset), fine-tuning corpus data/task (e.g., BC5), fine-tuning corpus size, number of tasks/datasets for performance evaluation, hardware type (e.g., GPU). These criteria were developed and iteratively improved over a pilot extraction round conducted by two authors (SED and BVI).

We summarized our findings in a narrative fashion bolstered by descriptive statistics of extracted parameters.

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<sup>1</sup> <https://harzing.com/resources/publish-or-perish>.



**Fig. 1** **A** Number of included articles over the chosen time frame. **B** Top 10 sources of included articles and their frequency, including NLP venues and biomedical journals

## 2.6 Risk of bias assessment

We assessed the rigour of each study against three predefined criteria: (1) Is the process of splitting training from validation data described? (2) Are there methods described for avoiding overfitting or underfitting? (3) Does the dataset or assessment measure provide a possibility to compare to other tools in the same domain?

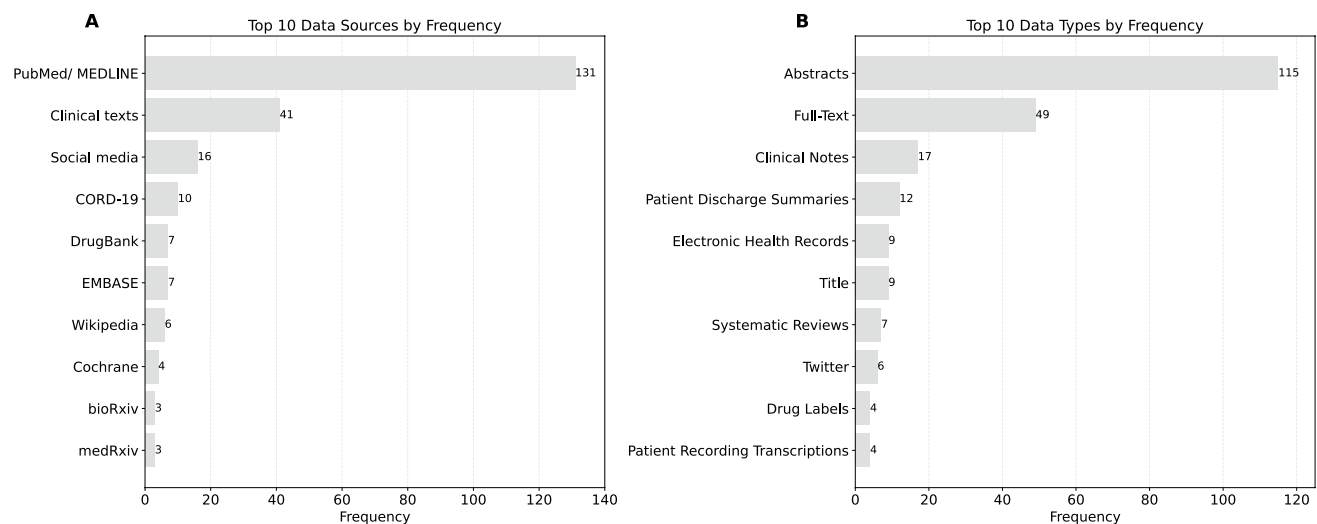
## 2.7 Changes from protocol

While our initial protocol aimed to evaluate all machine learning-based approaches for BioNLP, we decided to shift the focus specifically to the emerging field of LLMs. Originally, we wanted to include works published from 2014 onwards, representing the beginning of deep learning methods for automated text analysis, such as sequence to sequence learning [17]. To align with our new focus on LLMs, we updated our inclusion criteria to works published from 2017 onwards, marking the introduction of LLMs with the advent of the transformer architecture. We excluded predecessor sequence-based approaches like recurrent neural networks and long short-term memory networks because they do not exhibit the scale and capabilities characteristic of LLMs. This adjustment ensures that our evaluation is focused on the most current and impactful developments in the field.

## 3 Results

### 3.1 Eligible publications

In total, 13,823 publications were retrieved from our database search. 721 publications were eligible for full-text screening (see Appendix Fig. 9). A total of 196 articles were included in our final review, with 137 coming from biomedical journals and 59 from the NLP conference outputs (Fig. 1A). Figure 1B also shows the key sources of these articles, highlighting contributions from acknowledged NLP venues such as the ACL/Findings and the EMNLP conferences, as well as biomedical journals such as BMC Bioinformatics and the Journal of Biomedical Informatics.



**Fig. 2** Frequency of specific data sources (A) and data types (B) used for LLM development and testing. A paper could use more than one data source and type

### 3.1.1 Risk of bias of eligible articles

The overall risk of bias was moderate with 89% of studies reporting whether training and validation data were split, 25% of studies explicitly reported measures to mitigate over-/underfitting of models, and 84% of studies reporting on benchmarking their approaches against other methods.

## 3.2 Applications

### 3.2.1 Data sources and data types

Figure 2 introduces the top 10 data sources used to inform the training and refinement phases of the LLMs development. The most frequently reported ones were PubMed/Medline (131 publications, 58%), clinical texts (i.e., textual data generated in clinical settings such as electronic health records; 41, 18%), and social media (16, 7%).

The most commonly used data types were abstracts (115 publications, 50%) and full texts of scientific articles (49, 21%) as well as clinical notes (17, 7%) (Fig. 2).

### 3.2.2 Biomedical application domains

Eligible publications were grouped into 6 main categories of biomedical applications for LLMs (see Table 1). Those categories were aligned with commonly used terms in the biomedical NLP community and defined based on the types of data processed (e.g., clinical records, scientific literature, social media) and the intended use cases as reported in the papers [18, 19]. For instance, the primary goal of mining clinical texts is to improve healthcare outcomes, while literature-based discovery focuses on advancing knowledge in biology and medicine. The top three categories were literature-based discovery and knowledge extraction (73 publications, 38%), methodological advances in NLP (57, 29%), and drug discovery and pharmacovigilance (24, 12%) (Fig. 3A). Further details about applications within the top 4 most frequent domains are provided in the Appendix Fig. 10.

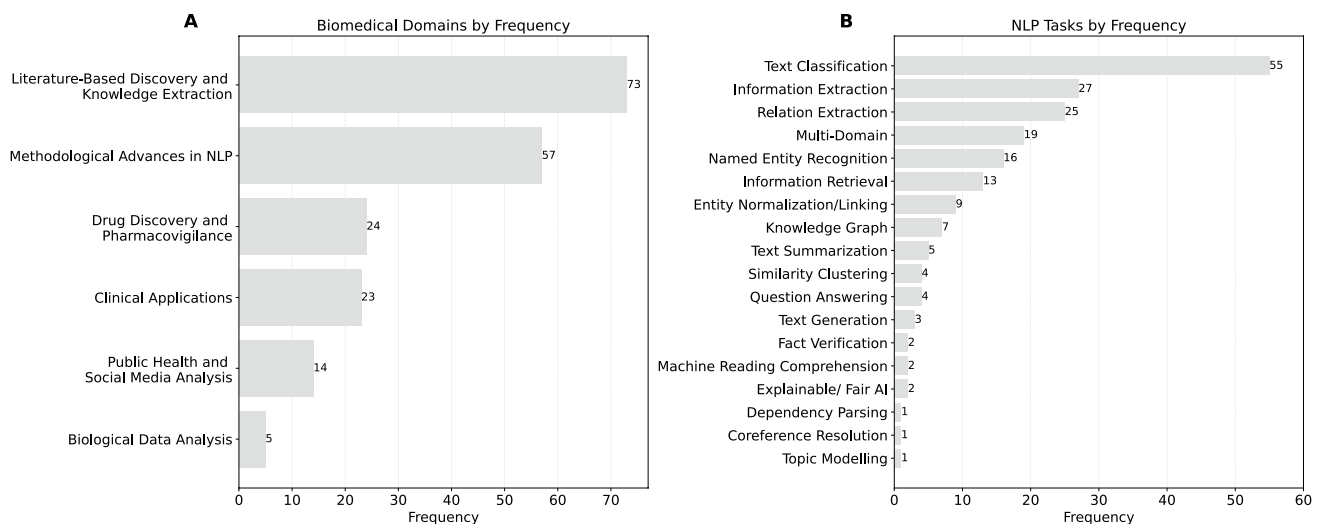
### 3.2.3 NLP tasks

A variety of NLP tasks has been employed by the eligible articles, most commonly text classification (55 publications, 28%), information extraction (27, 14%), and relation extraction (25, 13%) (Fig. 3B). Matching the main domains of application with the NLP tasks shows notable links (Fig. 4).

In literature-based discovery, LLMs have been used for two main applications: text classification and information retrieval. Text classification can streamline the organization of large literature volumes by assigning predefined categories

**Table 1** Main application domains for LLMs in biomedicine. Abbreviations: NLP, natural language processing

| Category                                            | Definition                                                                                                                   | Example paper |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------|
| Clinical applications                               | NLP applications directly related to patient care, clinical decision support, and electronic health records                  | [20]          |
| Literature-based discovery and knowledge extraction | Techniques for discovering new knowledge by mining scientific literature, including summarization and information extraction | [21]          |
| Drug discovery and pharmacovigilance                | NLP methods applied to the development of new drugs, detection of adverse drug reactions, and monitoring of drug safety      | [22]          |
| Biological data analysis                            | Applications focusing on understanding biological processes, including genomic and proteomic data interpretation             | [23]          |
| Public health and social media analysis             | Extracting health-related information from media and social networks to monitor trends, outbreaks, and public sentiment      | [24]          |
| Methodological advances in NLP                      | Development of new or improved NLP methodologies, algorithms, or tools applicable to biomedical data                         | [25]          |



**Fig. 3** Biomedical domains (A) and NLP tasks (B) by article frequency. Each paper was assigned to a single primary application domain and one main NLP task

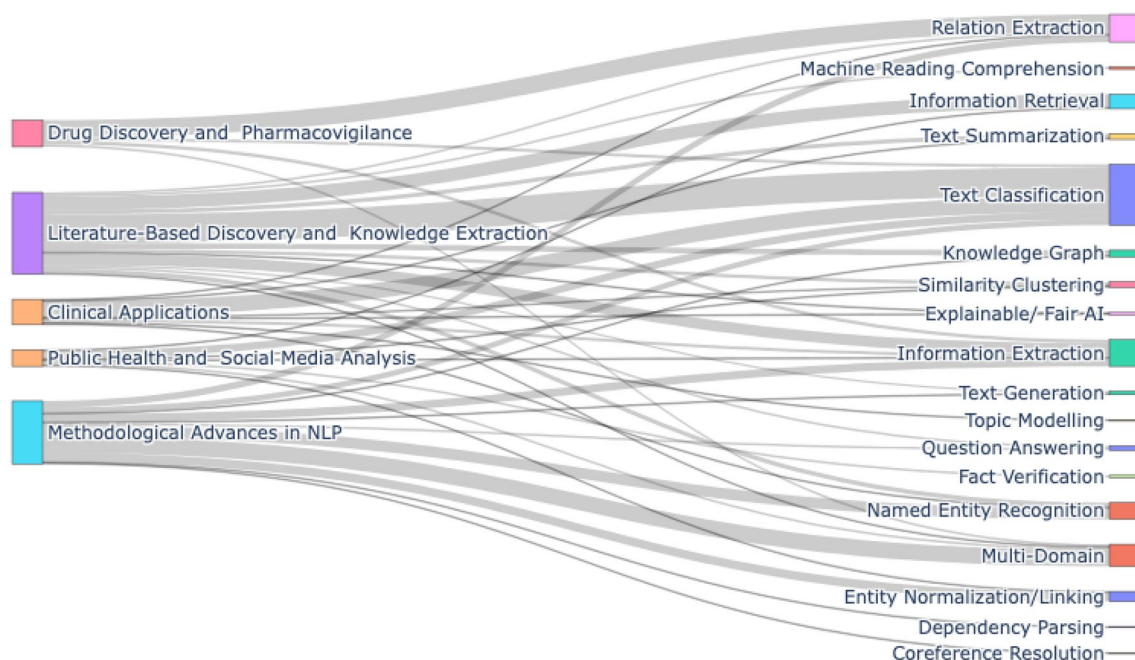
or labels to documents. This approach has been effectively applied to enhance reference prioritization in systematic reviews [26–30]. Text classification has also been used to better understand the structure of papers, for example by automatically predicting sections and headers in Electronic Health Records [31]. Information retrieval focuses on enabling users to find and access the specific information they need from among vast amounts of available content. Various applications have been developed to improve the discovery of novel insights by more effectively aggregating data from existing knowledge databases, as well as improving article recommendation service [32, 33]. For instance, pubmedKB serves as a web server designed to extract and visualize semantic relationships between genes, diseases, chemicals, and variants within PubMed abstracts [34].

Methodological advances in NLP span various tasks, with multi-domain analysis standing out. This area focuses on designing methods or frameworks capable of addressing multiple NLP tasks through a unified strategy. An illustration of this is BioGPT, a generative transformer model, which, after being pre-trained on an extensive biomedical literature corpus, demonstrates its versatility across six different BioNLP challenges [35]. Named entity recognition and entity normalization are also challenges commonly tackled together. For example, [36] developed a multi-task model that simultaneously learns sentence-level and token-level labels for NER, utilizing BioBERT for text encoding and sharing hidden states across tasks. [37] introduced an architecture that enhances pre-trained language models by integrating domain-specific knowledge from diverse sources such as the UMLS Metathesaurus and Wikipedia articles on diseases, through knowledge adapters and an attention-based controller, demonstrating performance improvements on various biomedical NLP tasks.

The pharmacological domain utilizes relation extraction the most. This task involves identifying and extracting relationships between entities such as drugs, genes, or proteins from text, which is crucial for pharmacological research and drug development. For example, [38] utilize different BERT models for drug-drug interaction extraction by integrating drug representations from a pharmaceutical knowledge graph and corpus text. [39] used contextual information in extracting long-distance adverse drug events from clinical notes with BERT models. There were also overlaps between domains, such as work that introduced a system for detecting adverse drug reactions from text on social media by fine-tuning BERT [40].

In the clinical applications domain, text classification is frequently utilized for predicting patient outcomes. For instance, in-hospital mortality predictions have been made by combining time series data from various medical devices with clinical notes found in electronic health records [41, 42]. Additionally, there are applications in mental health, such as the automated detection of mental conditions using transcribed patient recordings [43, 44].

Finally, social media platforms have been utilized for mental health screening, employing text classification to identify suicidal risk and predict mental health disorders from user-generated content, such as Reddit and Twitter posts [45–47]. Additionally, this NLP task has been applied for detecting nuanced emotional states within online health communities [48]. Another recent development has been the use of fact-verification techniques to authenticate statements related to COVID-19 [49, 50].



**Fig. 4** Sankey Diagram showing the relationships between the biomedical domains and the utilized NLP applications. Each domain is represented as a source node, while the associated NLP applications are shown as the target nodes. The thickness of the flows between the two is proportional to the number of articles that exhibit this connection

### 3.3 Large language models

#### 3.3.1 Models overview and trends over time

The most prominently used LLMs were encoder-based BERT models (275 models, 91%), followed by generative models such as GPT and T5 (27, 9%) (Fig. 5).

Figure 5 presents the annual usage trends of the top five LLMs. While BERT models are prominently used in earlier years, GPT models see increasing use in 2023. The data also shows that general-purpose models such as BERT are replaced with more specialized models, e.g., BioBERT and SciBERT. Studies also tended to compare more models over time, see Appendix Fig. 11.

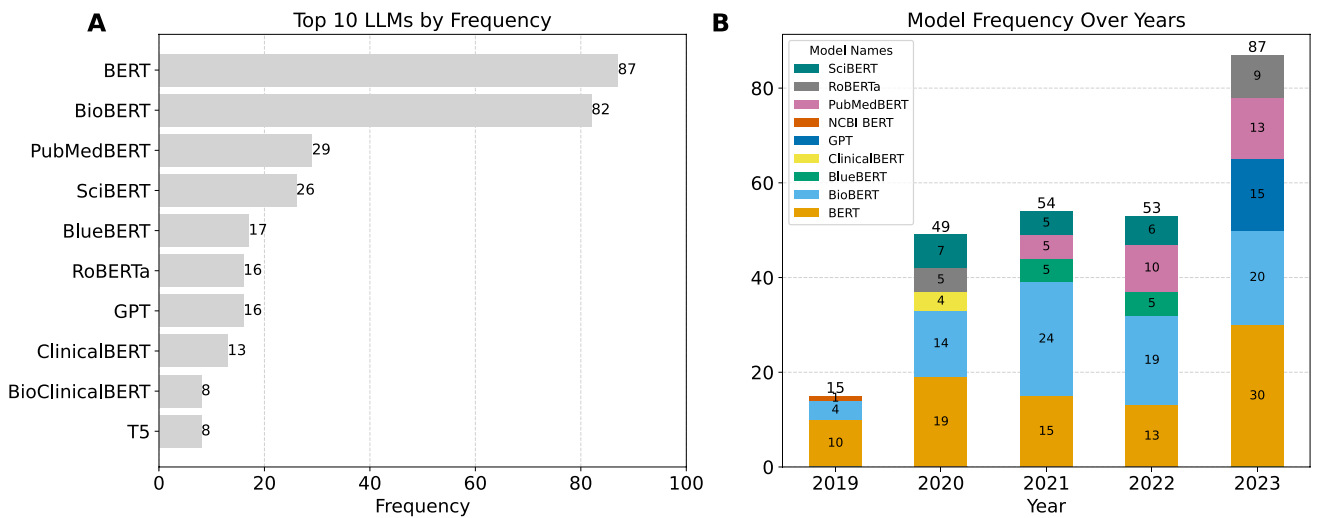
The original BERT model is pre-trained on a large, diverse corpus of text from sources like Wikipedia and BooksCorpus, making it versatile for a wide range of NLP tasks across various domains. It captures general language patterns and can be fine-tuned for specific tasks such as sentiment analysis or question answering [51].

Biomedical BERTs, such as BioBERT or ClinicalBERT, are variants of BERT that have been either further pre-trained, or trained from scratch, on biomedical-specific texts, including scientific literature, clinical notes, and medical databases like PubMed. This additional training enables the model to understand and generate text with the specialized vocabulary, concepts, and context unique to the biomedical field [52]. Table 2 provides an overview of the main BERT-based models encountered in literature and their training corpora.

The research involving the emerging GPT technology also worked with variations of the models. The conversational interface ChatGPT, was the most frequently cited model (6 publications). GPT-3 or GPT-3.5 were collectively mentioned in 5 publications. The specialized model BioGPT appeared in 3 publications, while GPT-4 was referenced in 2 publications [35].

### 3.4 Technical setup: hardware and programming languages

The most commonly employed hardware was GPUs (92%) (Fig. 6). Of note, there was a lack of reporting on hardware use in many instances (118 publications).



**Fig. 5** **A:** Most frequently used models. **B:** Top 5 models utilized each year. Each paper could use more than one LLM

**Table 2** Training corpora for common BERT-based models

| Training corpus  | BERT [51] | BioBERT [53] | PubMedBERT [52] | SciBERT [54] | BlueBERT [55] |
|------------------|-----------|--------------|-----------------|--------------|---------------|
| General          | ✓         | ✓            | ✗               | ✗            | ✓             |
| PMC              | ✗         | ✓            | ✓               | ✗            | ✗             |
| PubMed           | ✗         | ✓            | ✓               | ✗            | ✓             |
| Semantic scholar | ✗         | ✗            | ✗               | ✓            | ✗             |
| Clinical notes   | ✗         | ✗            | ✗               | ✗            | ✓             |

“General” refers to diverse text sources, “PMC” refers to full-text articles from PubMed Central, “PubMed” refers to biomedical literature abstracts, “Semantic Scholar” includes full-text paper from scientific literature (also non-biomedical), and “Clinical Notes” are de-identified medical records

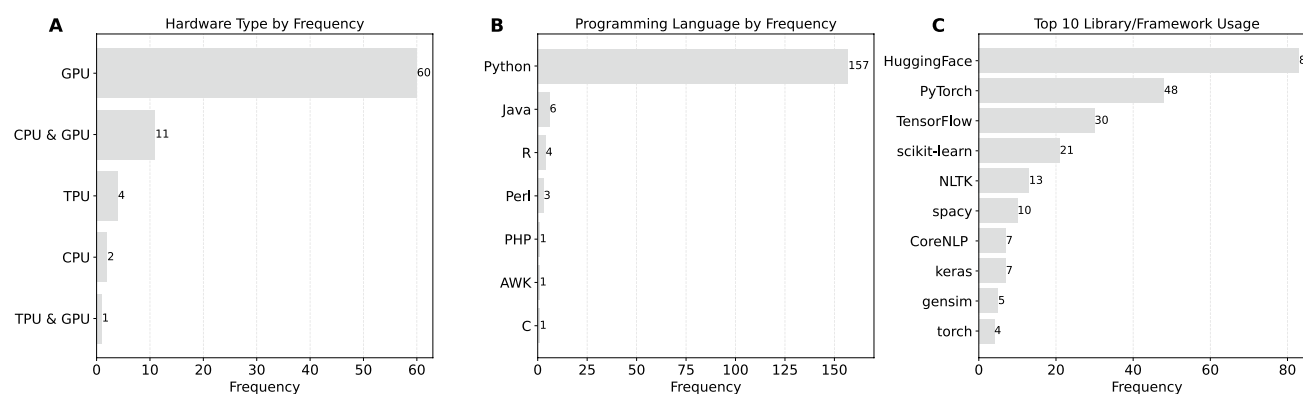
The vast majority of studies used the Python programming language for their technical setup (157 publications, 91%) (Fig. 6). JAVA, R, Perl, PHP, AWK, and C were rarely used. HuggingFace (83 publications, 36%), PyTorch (48, 21%), and TensorFlow (30, 13%) were the most commonly reported computational libraries for using LLMs (Fig. 6). Some studies used more specialized libraries such as Stanford CoreNLP, spaCy, Keras, NLTK, and Torch. Notably, many studies (37) did not report the programming language and computational library used.

### 3.5 Fine-tuning tasks and datasets

#### 3.5.1 Benchmark datasets

The field of BioNLP relies heavily on the availability of standardized datasets and benchmarks to assess and compare the performance of language models. Many of the reported datasets are collaborative efforts released for competitions and shared tasks, and designed to facilitate fair comparisons among different research teams. An overview of the most common datasets is provided the Appendix Table 3. Key competitions and shared tasks that emerged from the eligible studies include:

- Informatics for Integrating Biology and the Bedside (i2b2) Challenges: The i2b2 Center, active from 2004 to 2014 and funded by the NIH, focused on releasing clinical records for NLP research. Since 2018, this initiative is officially known



**Fig. 6** Reported technical setup regarding hardware (A), programming language (B), and computational library (C) used for the method implementation. Each paper could use more than one programming language and library. Not all of the libraries are specific to LLMs, but were rather reported to support data transformation and development of baseline algorithms

as n2c2<sup>2</sup>. The most frequently used dataset from this area is the 2010 i2b2/VA for concepts, assertions, and relation extraction. Other datasets mentioned in the reviewed papers include the 2012 corpus annotated for events, temporal expressions and temporal relations [56], and the de-identification datasets of i2b2 2014 [57]. Three studies utilized data from the 2019 n2c2/OHNL track on clinical semantic textual similarity [58–60], and one study tackled the 2022 n2c2 challenge, focusing on medication extraction and event/context classification [61].

- **BioCreative (BC) Workshops:** These workshops present challenges aimed at extracting and annotating biological entities and their relationships from scientific literature. Prominent datasets associated with this venue, and commonly used in the reviewed papers, include BC5, which addresses disease and chemical annotations, and BC4, which focuses on chemical annotations. Additionally, datasets like LitCovid and DrugProt were also frequently utilized in the reviewed studies [38, 62–64].
- **Text REtrieval Conference (TREC):** TREC is an ongoing series of workshops that cover a wide range of information retrieval topics. Datasets associated with TREC, such as TREC-COVID and Health Misinformation, have been employed in four of the reviewed research papers [65–68]. One paper also utilized three datasets of the TREC series 2017–2019 for clinical decision support [69].
- **BioNLP Shared Task (BioNLP-ST):** This series is aimed at advancing the extraction of detailed bio-molecular events from scientific texts. The provided data from different years has been utilized nine times in the reviewed papers. For example, the 2011 Protein Coreference dataset is used for the evaluation of methods for coreference extraction among protein/gene [70]. A combination of the datasets from BioNLP 2009, 2011 and 2013 is used in [71] to benchmark a novel fine-tuning approach.

### 3.5.2 Custom-developed datasets

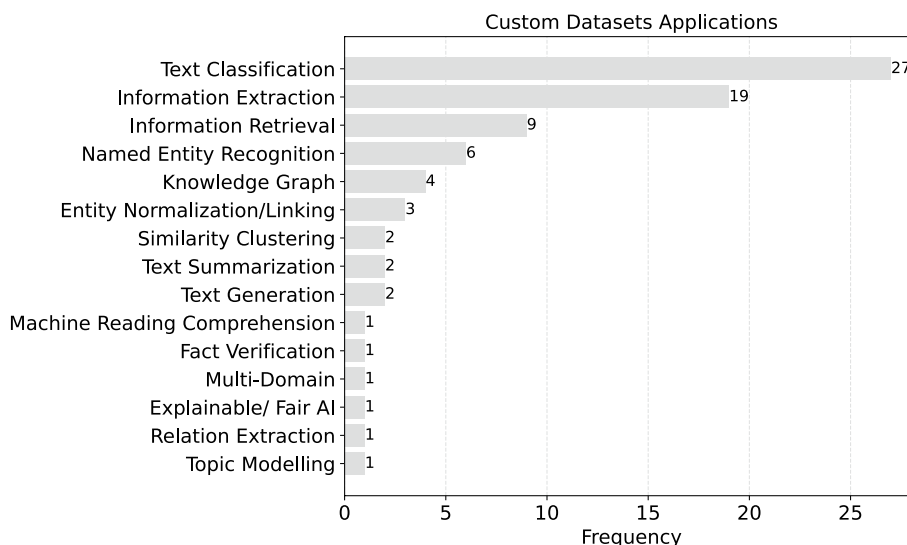
41% of the publications reported on the development of new datasets. These datasets were mostly in the area of text classification, information retrieval/extraction, and named entity recognition (Fig. 7).

## 3.6 Transparency of methods

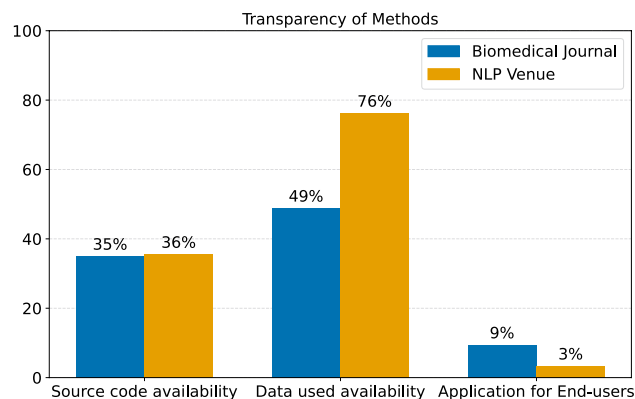
There was a moderate level of code (e.g., algorithms and computational methods transparency) with fewer than 40% of the studies making their code available alongside the publication. While over three-quarters of studies in NLP venues used a publicly available dataset, only 49% of studies in biomedical journals shared their data. Finally, only a minority of studies provided end-user applications, with these being more commonly shared in biomedical journals compared to NLP venues (Fig. 8).

<sup>2</sup> <https://portal.dbmi.hms.harvard.edu/projects/n2c2-nlp/>.

**Fig. 7** Target applications for which custom data has been annotated or collected



**Fig. 8** Transparency of LLM methods: source code and data availability, and end-user application deployment



## 4 Discussion

### 4.1 Main findings

The objective of our scoping review was to explore and organize the specific tasks in BioNLP that are being tackled with LLMs. In the biomedical domain, LLMs have found applications across a diverse set of tasks, including knowledge management, drug discovery, clinical applications, and synthesizing evidence. Within these areas, tasks such as text classification, relation extraction, and named entity recognition/information extraction stand out for their prominence. The primary sources of data for these studies are often PubMed/Medline abstracts or clinical documents, such as electronic health records. While BERT-based models continue to be widely used, there has been a noticeable surge in the adoption of GPT-based models since 2023. Moreover, the technical frameworks for these studies typically involve the Python programming language, with HuggingFace and PyTorch being the most popular computational libraries.

#### 4.1.1 Limitations and challenges of LLMs for BioNLP

Most LLM applications reviewed in the present study were developed using information from scientific abstracts. This preference might be due to the easy access to and the concise nature of this data type. However, information from abstracts might not be well aligned with the corresponding full reports. For example, [72] analyse the state of reporting of primary biomedical research and find inconsistencies with respect to the reported sample sizes, outcome measures, result presentation and interpretation, and conclusions or recommendations between abstracts and full texts. This suggests the need to be cautious when developing applications that rely only on the information reported in abstracts,

especially for evidence-synthesis. Article abstracts and article bodies can differ not only in content, but also in their structural, linguistic, and semantic composition [73].

The most common text-based applications of LLMs in the biomedical field were within the knowledge management domain, showcasing their potential as a solution to the challenges brought on by the exponential growth of biomedical literature [74–78]. However, we also observed that LLMs have been mostly used as a supportive tool by assisting with preliminary tasks in biomedical research, but they have not yet been capable of replacing more nuanced and critical tasks entirely. For example, while LLMs can expedite the initial screening of literature for relevant studies and assist in the extraction of data from texts, they cannot replace the expert analysis required for interpreting study results, assessing the quality of evidence, or making clinical decisions [79, 80].

Generative models have been benchmarked against BERT-based approaches as a classification and information extraction technique for biomedical literature curation [81–83]. For information extraction tasks, fine-tuning domain-specific BERT models still generally achieves the best performance. The challenges when using LLMs like ChatGPT in this context include verification difficulty due to rephrased entities and relations, the production of plausible-sounding but factually incorrect information, and the challenge of finding appropriate prompts for tasks like NER. Closely related is the issue that most benchmark datasets are designed for supervised models that require structured predictions, which are hard to obtain from generative models [84–86].

Further considerations include the challenge of maintaining model accuracy amidst the fast pace of medical knowledge evolution, the models' limited interpretability, susceptibility to hallucinations and biases, and ethical and privacy concerns related to handling sensitive patient information. These issues emphasize the importance of cautious and responsible LLM implementation in the biomedical domain [7, 8, 87–89].

#### 4.1.2 Transparency of methods

A minority of studies present end-user applications, but a notable portion leverages openly available datasets or share their proprietary datasets. Despite this positive trend, only around one third of these studies include references to their code repositories, a critical factor for ensuring the validity, reproducibility, and generalizability of their methods [90, 91]. While guidelines for reporting experimental setups and results exist in the NLP and machine learning fields, there seems to be a need for further approaches to improve the reporting rates [92–94]. Moreover, the use of API-based models such as GPT introduces additional challenges for research transparency and reproducibility [95]. Shifting towards open-source models can address these issues by providing full access to the models' codebase, fostering an environment where research findings can be easily replicated, scrutinized, and extended by the wider scientific community.

Recently, the landscape of generative biomedical models has been enriched by open-source solutions such as LLama 2, MEDITRON, and BioMistral [96–98]. These models are rapidly narrowing the performance gap with proprietary counterparts, while also promoting a more transparent, reproducible, and independent research ecosystem. Furthermore, the development of streamlined versions, such as *ollama*, optimized for local computing resources, highlights this trend by enhancing the practicality and accessibility of LLMs for a broader range of research applications<sup>3</sup>.

#### 4.1.3 Current developments

A significant advancement in LLMs is the shift from being designed specifically for single tasks, such as text classification, to the development of general-purpose capabilities applicable across a wide range of tasks [99, 100]. Current advances in this area suggest a movement towards a “generalist biomedical AI” [99, 101]. This innovation is set to transform healthcare by enabling: (1) users to customize models for new tasks using plain language; (2) models to process and generate outputs using diverse data types, such as images, text, and laboratory results; and (3) models to access extensive medical knowledge, thereby enhancing their ability to reason and provide explanations in medically precise terms [102, 103]. Below are two promising applications of those AI models, which mark new possibilities in the field.

Generalist LLMs can help to streamline the analysis of medical records, images and data from patient monitoring devices, thus enabling healthcare professionals to access critical information more rapidly [104, 105]. For example, by integrating patient history, current metrics and clinical knowledge, such AI models could provide patient status summaries, early warnings, as well as recommendations for future care [99]. In addition, those models can assist in generating

<sup>3</sup> <https://ollama.com/>

medical summaries by monitoring electronic patient information and clinician-patient conversations. Based on the collected inputs, they could automatically create documents like electronic notes and discharge reports, allowing clinicians to review and approve documents with reduced administrative burden [99]. Summarization is also critical for patient communication, where LLMs could translate complex medical jargon into simpler language, making it easier for patients to understand their conditions and treatment plans [106].

In education, LLMs offer new learning opportunities by interacting conversationally, answering follow-up questions, and providing tailored insights. Early applications demonstrate the potential of these models for personalized learning and teaching assistance, as well as augmentation of medical procedures [99, 107, 108]. LLMs can also help educate patients about medical procedures, risks, and clinical care at various literacy levels and in multiple languages [108]. With the ability to interpret heterogeneous types of data, such as unstructured descriptions of symptoms or patient-provided medication logs, the model could provide detailed advice and explanations [99].

## 5 Limitations

First, our literature review primarily focuses on text-based applications of LLMs in biomedicine. This emphasis may not fully capture the multimodal nature of biomedical data, which includes images, genomic sequences, and other non-textual information. However, we believe that textual data remains an underutilized resource in the biomedical field. The vast amounts of unstructured text, such as clinical notes, research articles, and electronic health records, offer opportunities for LLMs to advance knowledge extraction, information retrieval, and decision support.

Second, the categorization of application domains for large language models was based on our manual assessment. While we tried to align the labels with more commonly used terms. However, we acknowledge that it may still fall short of fully capturing the diversity of the field, with some overlap between categories. Future research could benefit from a more standardized or nuanced framework to better reflect the wide range of LLM applications in biomedicine.

Finally, we omitted articles published in languages other than English, which could cause language bias.

## 6 Conclusion

This scoping review provides an overview of how LLMs are used for NLP tasks in biomedicine and health sciences. The findings from our review underline the rapid progress of LLMs, emphasizing their potential in accelerating discovery and enhancing health outcomes. These advancements signal a promising avenue for leveraging LLMs in biomedicine and health, given their capacity to process and analyze complex biomedical texts with high proficiency. However, we also acknowledge the inherent risks and challenges associated with the deployment of LLMs such as ChatGPT in these sensitive areas, including evidence synthesis. Issues such as the generation of fabricated information and concerns over legal and privacy implications pose significant hurdles to the safe and ethical application of these technologies. These challenges highlight the need for careful consideration and management to mitigate risks associated with the use of LLMs in biomedicine and health.

**Acknowledgements** We thank Alisa Berger for conducting the comprehensive literature search.

**Author contributions** Conception and design of study: SED, BS, BVI; Acquisition of data: SED, SQ, BVI; Analysis of data: SED; Interpretation of data: SED, SQ, BS, TE, J-PG, GS, BVI; Drafting the initial manuscript: SED; All authors critically revised the paper draft.

**Funding** Swiss National Science Foundation (No. 407940-206504, to BVI) UZH Digital Entrepreneur Fellowship (No number, to BVI). The sponsors had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

**Data and code availability** All data and code that support the findings of this study are available in a public GitHub repository (<https://github.com/Ineichen-Group/LiteratureReview-LLM-Biomedicine/tree/main>). For any questions regarding the data, meta-data, or analysis code, contact the corresponding author, SED.

## Declarations

**Competing interests** The authors declare no competing interest related to this study.

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## Appendix

### A: Search string

Supplementary search string ('data mining'/exp OR ((data NEXT/1 mining):ti,ab,kw)) AND (literature:ti,ab OR abstract:ti,ab OR abstracts:ti,ab OR text:ti,ab OR articles:ti,ab) OR (((analyz\* OR analys\* OR extract\* OR screen\* OR evaluat\* OR classif\* OR 'natural language processing') NEAR/3 (literature OR abstract OR abstracts OR text OR articles)):ti,ab,kw) OR ((text NEXT/1 mining):ti,ab,kw) AND 'natural language processing'/exp OR 'artificial neural network'/exp OR 'support vector machine'/exp OR 'machine learning'/de OR 'automated pattern recognition'/exp OR 'artificial intelligence'/de OR 'semi automation':ti,ab,kw OR automation:ti,ab,kw OR 'artificial intelligence':ti,ab,kw OR ai:ti,ab,kw OR 'natural language processing':ti,ab,kw OR ((machine NEXT/1 (intelligence OR learning)):ti,ab,kw) OR (((('text mining' OR 'data-mining') NEXT/3 (tool\* OR technique\* OR system)):ti,ab,kw) OR (((deep OR comput\* OR model\* OR convolutional OR artificial OR algorithmic OR connectionist OR mathematical) NEXT/1 'neural network\*'):ti,ab,kw) OR ((ann NEXT/3 (analysis OR approach OR method\* OR model\* OR output OR technique\* OR training\*)):ti,ab,kw) OR ((connectionist NEXT/1 (model OR network)):ti,ab,kw) OR (('support vector' NEXT/1 (machine\* OR classif\* OR network OR regression)):ti,ab,kw) OR ((auto-mated NEXT/3 (tool\* OR technique\* OR system OR 'pattern recognition' OR analyz\* OR analys\* OR extract\* OR screen\* OR evaluat\* OR classif\*)):ti,ab,kw)

### B: Flow diagram of the excluded and included studies

Appendix Figure 9 illustrates the process of the scoping review, showcasing the flow of information through the different phases of a literature review, including identification, screening, eligibility, and inclusion of studies.

### C: Target biomedical applications subdomains

Further details about downstream applications within the top 4 most represented biomedical domains are provided in Appendix Fig. 10. These subcategories have been identified based on the objectives outlined in the reviewed papers.

### D: Number of tested models per paper over time

The mean number of LLMs used by individual papers ranges between 1.36 and 2.39, with a slight upward trend over time (Appendix Fig. 11).

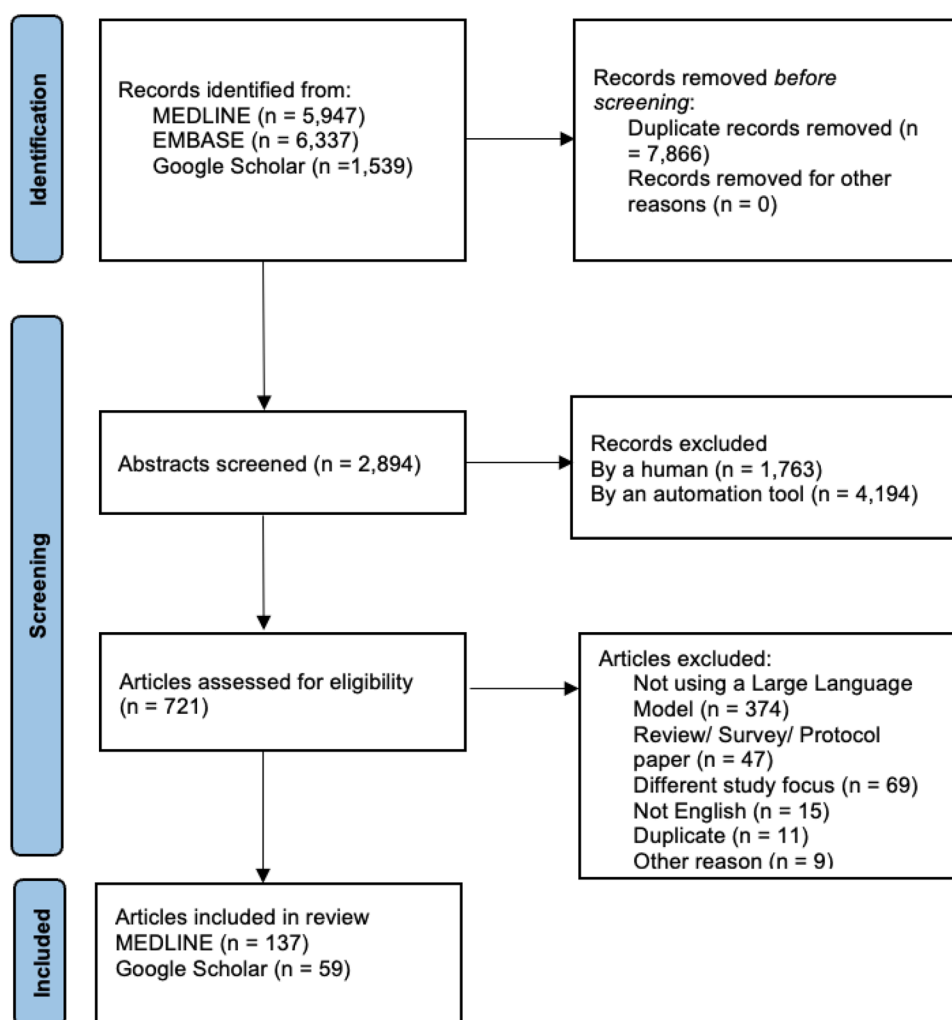
### E: High-level LLM background

The GPT-like and BERT-like models represent two distinct LLM approaches, differentiated by their architecture, training methods, and use cases [110].

#### BERT-style language models

BERT employs a bidirectional framework, analyzing text in both directions simultaneously, which is enabled by its use of the Transformer encoder architecture. This bidirectionality allows BERT to understand the context of a word based on its entire surrounding text, making it good at tasks that require a deep understanding of language context, such as sentiment analysis, question answering, and named entity recognition. BERT's pre-training involves masked language

**Fig. 9** PRISMA Flow Diagram [109]



modeling and next sentence prediction, tasks that help it learn a comprehensive understanding of language structure and flow.

While most reported models follow a BERT-based architecture, they differ in the pretraining corpus used. The standard BERT [51] model is pretrained on texts from Wikipedia and BookCorpus, which is considered general-domain. To improve its performance in BioNLP tasks, the models can be trained on biomedical text corpus in two ways [52]:

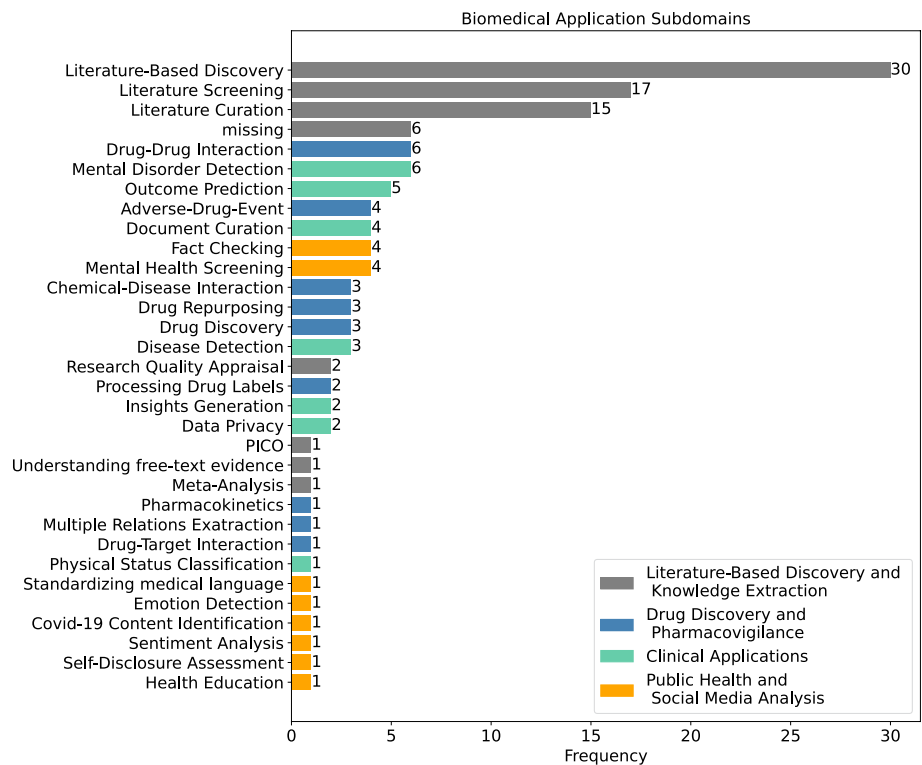
1. **Mixed-Domain Pretraining:** Starting with a general-domain BERT model training is continued using biomedical texts. In the case of BioBERT [53] this includes PubMed abstracts and PubMed Central (PMC) full-text articles. BlueBERT [55] uses both PubMed abstracts and de-identified clinical notes from MIMIC-III [111].
2. **Domain-Specific Pretraining:** In this setup the language model is trained using purely in-domain data. PubMedBERT [52] follows this approach and is pretrained from scratch on abstracts and full-texts from PubMed only.

SciBERT [54] is also trained from scratch on a purely scientific corpus from Semantic Scholar. However its pretraining corpus is a mixture of biomedical and computer science texts.

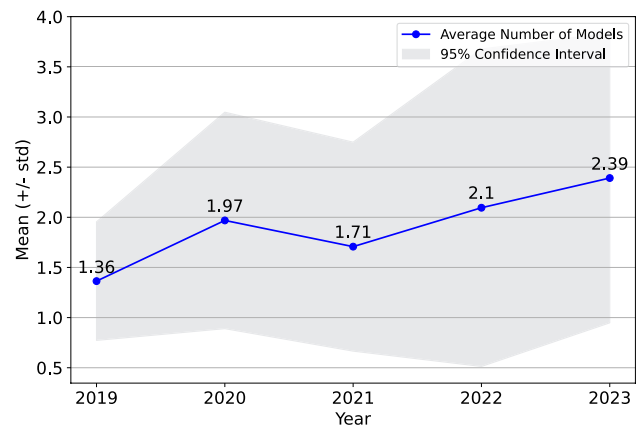
### GPT-style language models

GPT-like models operate on a unidirectional framework, processing text from left to right and utilizing the Transformer decoder architecture, making it well fit for generative tasks such as text completion and content creation. Its training is focused on predicting the next word in a sequence, optimizing the model for generating coherent and contextually

**Fig. 10** Number of articles assigned to each subdomain



**Fig. 11** Average number of different models used per paper each year



relevant text. While BERT-like architectures typically require task-specific fine-tuning to achieve optimal performance, models like GPT-3 demonstrate few-shot and zero-shot learning capabilities. Few-shot learning refers to the model’s ability to perform tasks with a very limited amount of training data, while zero-shot learning refers to its ability to perform tasks without any task-specific training data. This is achieved through advanced prompting techniques and in-context learning, where the model generates responses based on the context provided in the prompt [112].

**F: Overview of common research datasets**

Appendix Table 3 provides a compilation of the most commonly used benchmarks according to the reviewed papers. Named Entity Recognition (NER) datasets such as NCBI-disease, BC5-disease, and BC2GM were frequently utilized, with

**Table 3** Overview of commonly used datasets for different NLP tasks

| Task                | Dataset                  | Frequency | Annotation type                              | Text type                        | Size   |
|---------------------|--------------------------|-----------|----------------------------------------------|----------------------------------|--------|
| NER                 | NCBI-disease [113]       | 21        | Disease                                      | Abstract                         | 793    |
|                     | BC5-disease [114]        | 14        | Disease                                      | Abstract                         | 1500   |
|                     | BC2GM [115]              | 13        | Gene/Protein                                 | Sentence                         | 20,000 |
|                     | BC5-chem [114]           | 12        | Chemical                                     | Abstract                         | 1500   |
|                     | JNLPBA [116]             | 9         | Protein, DNA, RNA, cell line                 | Abstract                         | 2404   |
|                     | BC4-chem [117]           | 6         | Chemical                                     | Abstract                         | 10,000 |
|                     | LINNAEUS [118]           | 5         | Species                                      | Full text                        | 100    |
|                     | Species-800 [119]        | 5         | Species                                      | Abstract                         | 800    |
| Relation extraction | ChemProt [120]           | 10        | Chemical-protein interactions                | Abstract                         | 31,784 |
|                     | DDIExtraction-2013 [121] | 10        | Drug-drug interaction                        | Texts and Abstract               | 1017   |
|                     | BC5-relations [114]      | 9         | Chemical-disease interaction                 | Abstract                         | 1500   |
| Question answering  | BioASQ [122]             | 6         | Question-answer pairs and supporting context | Questions with related documents | 4719   |
| Multiple            | MIMIC-III [111]          | 8         | Multi-label                                  | EHR                              | 53,423 |
|                     | EU-ADR [123]             | 5         | Drug, disorder, targets                      | Abstract                         | 100    |
|                     | 2010 i2b2/VA [124]       | 5         | Multi-label                                  | Clinical reports                 | 1748   |
|                     | Systematic reviews       | 5         | Multi-label                                  | Full text                        | n.a.   |

The “Frequency” column contains the number of papers that have used the dataset for fine-tuning or testing

NCBI-disease cited by 21 papers, emphasizing its importance in disease annotation from abstracts. Relation extraction datasets like ChemProt and DDIExtraction-2013 focused on chemical-protein and drug-drug interactions, showing usage with 10 mentions each. Additionally, multipurpose datasets like MIMIC-III, EU-ADR, and the 2010 i2b2/VA dataset were employed across various tasks, demonstrating their versatility in handling multi-label annotations from electronic health records (EHRs) and clinical reports.

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